

Sienna White

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EXPERIENCE & EDUCATION

University of California, Berkeley, Berkeley, California

Aug 2022 – Jan 2027

Ph.D. Candidate, Environmental Engineering

My dissertation research investigates harmful algal blooms in the Sacramento-San Joaquin Delta, supervised by Prof. Mark Stacey. My work integrates estuarine physics, numerical modeling, remote sensing, and inverse methods for biogeochemical and hydrodynamic models.

M.S., Environmental Engineering (2023)

I worked with Prof. Tina Chow to produce high-resolution wildfire smoke analysis fields by assimilating observational data into HRRR-Smoke model results.

Delta Stewardship Council, Sacramento, California

Aug 2025 – Jan 2026

Delta Science Collaboratory

Invited to participate in the Delta Science Collaboratory as an expert in harmful cyanobacteria algal blooms (C-HABs). Worked with a small team to produce a white paper detailing opportunities for future collaborative science in the Sacramento-San Joaquin Delta.

San Francisco Estuary Institute, Richmond, California

Sep 2020 – Aug 2022

Environmental Analyst

Advanced hydrodynamic + biogeochemical numerical models of San Francisco Bay to inform management decisions for the greater Bay Area. Specific emphasis was placed on investigating and validating our representation of nutrient cycling and primary production.

Fulbright Fellow, Technische Universiteit Delft, the Netherlands

Sep 2019 – Sep 2020

Researcher in river morphology.

Conducted a geospatial analysis to quantify morphological changes from a shipping infrastructure project in the Waal River.

Deltares, Delft, the Netherlands

Jun 2018 – Sep 2018

Visiting researcher

Developed physically-based roughness predictors for the Rijn River hydrological model.

Stanford University, Stanford, California

Sep 2015 – Jun 2019

B.S. with Honors in Environmental Engineering

Thesis: A Numerical Model for Sediment Flux and Flocculation in San Francisco Bay.

Environmental Fluid Mechanics Lab: Performed statistical analysis of sediment budgets for San Francisco Bay. Organized fieldwork in San Francisco Bay, deploying ADVs, ADCPs, RBRs, and thermistors.

TEACHING

Teaching Assistant, Environmental Fluid Mechanics (UC Berkeley),

Aug 2024 – Dec 2024

Teaching assistant for Environmental Fluid Mechanics, one of the core graduate classes in the Environmental Engineering masters program. I led weekly recitations, developed lecture notes, held office hours, and assisted with other teaching & grading tasks.

Lead Instructor, Environmental Science (Mt. Tamalpais College)

Jan 2024 – May 2024

Developed & taught an Environmental Science course at Mt. Tamalpais College, an accredited college within San Quentin State Prison. I was the lead instructor for a five-person teaching team, and coordinated our curriculum development and overall course structure throughout the semester.

Teaching Assistant, (Stanford University)

Sep 2015 – Jun 2019

I graded and/or helped teach several courses during my undergrad, including Energy Options for the 21st Century (TA), Modeling Environmental Flows (grader), and Storycraft, a theater class on personal storytelling (TA).

AWARDS & SCHOLARSHIPS

Outstanding GSI Award

2024

Given to top 10% of graduate student instructors at Berkeley.

1st Place, Communicate Your Science Essay Competition

2026

Essay competition in science communication. My essay, "The Things We Do For Mud," was published here.

Computational Science Graduate Fellowship

2023-2027

Four-year fellowship awarded by the Department of Energy to students pursuing degrees in the computational sciences.

Early Career Presenter Award 2025

Given to best early career talk at the Interagency Ecological Program in Sacramento, CA.

Berkeley Fellowship 2022

Two-year fellowship awarded to incoming Ph.D. students at UC Berkeley.

Firestone Medal 2019

Given to top 10% of honors theses at Stanford.

1st place, Creative Nonfiction Prize 2017

Won Stanford's annual award for best creative nonfiction essay written that year.

MENTORSHIP

Tia Böttger

Aug 2024 – Aug 2025

Worked closely with Tia, a post-bac student at USGS, on a year-long joint project studying turbulence and cyanobacteria in the Stockton Channel.

Josh Mendelson & Will Fahy

Dec 2025 – Feb 2026

Mentored Josh and Will, M.S. students, on a geospatial analysis of assimilated PM_{2.5} data.

Femke Bisschop

Jan 2020 – Jun 2020

Supervised Femke's bachelor thesis "Effects of Fixed Beds and Bottom Groynes in the Rhine River."

ACADEMIC SERVICE

Seminar Coordinator

Aug 2025 – Jan 2027

Coordinator for Berkeley's environmental fluid mechanics seminar, and additionally serve as the Berkeley representative for the Western Coastal Collaborative seminar series.

Admissions Committee

Dec 2025 – Feb 2026

Served on the MS and Ph.D. admissions committee for Berkeley's environmental engineering program.

SKILLS

Python, Julia, MATLAB, C++, LINUX/UNIX systems, QGIS, coastal hydrodynamic modeling (incl. DFlow-FM, SCHISM), biogeochemical modeling, large-eddy simulations, high-performance computing, GIT/version control

PUBLICATIONS & PRESENTATIONS

PAPERS

White, S.R., Sugrue, R.A., Guillotte, L., James, E., Wang, H., Ahmadov, R., Thakur, N. and Chow, F.K. "Hourly PM_{2.5} Estimates across California from 2018 to 2023." ACS ES&T Air 3.4 (2026): 968-980.

Ylla Arbós, C., A. Blom, S. R. White, R. Patzwahl, and R. M. J. Schielen. "Large-scale channel response to erosion-control measures." Water Resources Research 60, no. 3 (2024): e2023WR036603.

SUBMISSIONS & PRE-PRINTS

White, S.R., Böttger, T., Bouma-Gregson, K., Preece, E., Stacey, M. "Daily dynamics of phytoplankton growth in a low-energy tidal river." Limnology & Oceanography [in revisions].

Böttger, T., White, S.R., Bouma-Gregson, K., Stacey, M. "Diurnal stratification and circulation in Delta dead-end channels: Physical observations of the Stockton Channel." San Francisco Estuaries & Watersheds Sciences [under review].

Sugrue, R., Nguyen, A., White, S.R., Mendelson, J., Balmes, J., De La Rosa, R. Chow, F., Thakur, N. "Sub-daily fine particulate matter exposure improves characterization of respiratory health risk." Nature Health Communications [under review].

TECHNICAL REPORTS

S.R. White, P. Mugunthan, A.T. King, F. Karimpour, T.E.C. Kraus, E.B. Stumpner, B.A. Bergamaschi, D.B. Senn. "Delta-Suisun Biogeochemical Model: Calibration and Validation (WY2016, WY2011)." San Francisco Estuary Institute. Richmond, CA. September 2021.

S.R. White, A. King, F. Karimpour, P. Mugunthan, D. Senn. "Nutrients in the Northern San Francisco Estuary: Transport, Cycling, and Forecasted Changes after Nutrient Load Reductions." San Francisco Estuary Institute. Richmond, CA. March 2021.

F. Karimpour, S.R. White, A. King, P. Mugunthan, D. Senn. "Delta Biogeochemical Model WY2016: Progress Update." San Francisco Estuary Institute. Richmond, CA. December 2019.

CONFERENCES

S.R. White, G. Egan, O. Fringer, M. Stacey. "How does uncertainty aggregate in sediment transport models with flocculation dynamics?," in *Physics of Estuaries & Coastal Seas*, Portland, Maine, Aug 2026.

S.R. White, R.A. Sugrue, L. Guillotte, E. James, H. Wang, R. Ahmadov, N. Thakur, and F.K. Chow. "Data Assimilation of Wildfire Smoke and Monitoring Network Data to Estimate Hourly PM_{2.5} Across California (2018–2023)," in *Berkeley Atmospheric Science Colloquium*, Berkeley, CA, Mar 2026.

S.R. White, T. Böttger, K. Bouma-Gregson, E. Variano, M. Stacey. "Phytoplankton Species Competition in Low-Energy, Diurnally Stratified Environments," in *Ocean Science Meeting*, Glasgow, Scotland, Feb 2026.

S.R. White, T. Böttger, K. Bouma-Gregson, E. Variano, M. Stacey. "Does stratification drive harmful algal blooms in the Stockton Deep Water Channel?," in *State of the San Francisco Estuary*, Oakland, CA, Oct 2025.

S.R. White, T. Böttger, K. Bouma-Gregson, E. Variano, L. Lucas, M. Stacey. "How does vertical mixing influence phytoplankton in a low-energy tidal river?," in *Gordon Research Conference: Coastal Processes*, New London, CT, Jun 2025.

S.R. White, T. Böttger, E. Variano, K. Bouma-Gregson, L. Lucas, M. Stacey. "Vertical mixing in dead-end channels of the Delta," in *Interagency Ecological Program*, Sacramento, CA, Mar 2025.

S.R. White, T. Böttger, E. Variano, L. Lucas, M. Stacey. "Simulating algal species competition in the Sacramento-San Joaquin Delta using an idealized one-dimensional model," in *Physics of Estuaries & Coastal Seas*, Bordeaux, France, Sep 2024.

S.R. White, P. Mugunthan, A. King, F. Karimpour, D. Senn, "Exploring primary production and nutrient cycling in the Delta using a coupled hydrodynamic-biogeochemical model," in *California Water and Environmental Modeling Forum*, Folsom, California, Apr 2023.

S.R. White, A. Blom, "Quantifying sediment transport and associated bed elevation change over the fixed layers at Nijmegen and St. Andries," in *Netherland Centre for River Studies 2020*, Nijmegen, the Netherlands, Feb 2020.

S.R. White, O.B. Fringer, "Using D-Water Quality and D-Flow FM to model cohesive sediment transport in San Francisco Bay ," in *Delta Science and Software Days 2019*, Deltares, Delft, the Netherlands, Nov 2019.

S.R. White, O.B. Fringer, "Estimating a Sediment Budget in South San Francisco Bay from limited streamflow data," in *Ocean Sciences 2018*, Portland, Oregon, USA, Jan 2018.

OTHER

Sienna White, "The Hoverflies are Props: Notes on Fredrik Sjöberg's Memoir." *Anthroposphere: the Oxford Climate Review*, Nov 2019.

OTHER WORK EXPERIENCE

Sitka Island Institute , Sitka, Alaska	2016
Worked as an audio engineer on a podcast centered on indigenous experiences of climate change in S.E. Alaska	
Senior Producer , Stanford Storytelling Project, Stanford, CA	2016 – 2019
Organized events with visiting poets, authors; contributed to the quarterly podcast "State of the Human."	
Community radio , Radio Boise, Boise, ID	2014 - 2020
Ongoing volunteer with Radio Boise, Boise's community radio station. Work included podcast development, music programming, & grant writing.	